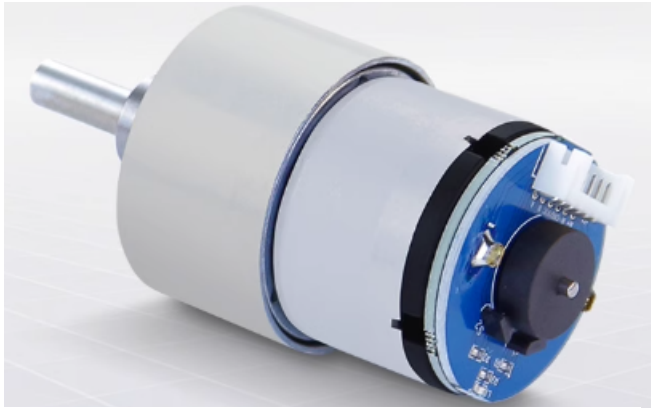


520 Motor Introduction and Usage

This article is used to explain the motor parameters, recommended supply voltage, and the recommended wiring method for this motor to be connected to the four-way motor driver board.

520 motor



Parameters	MD520Z19_12V	MD520Z30_12V	MD520Z56_12V	L-type 520 motor
Rated voltage	12V	12V	12V	12V
Motor type	Permanent magnet brush	Permanent magnet brush	Permanent magnet brush	Permanent magnet brush
Output shaft	6mm diameter D-type eccentric shaft	6mm diameter D-type eccentric shaft	6mm diameter D-type eccentric shaft	6mm diameter D-type eccentric shaft
Stall torque	3.1kg·cm	4.8kg·cm	8.3kg·cm	10kg·cm
Rated torque	2.2kg·cm	3.3kg·cm	6.5kg·cm	4.4kg·cm
Speed before deceleration	11000rpm	11000rpm	12000rpm	12000rpm
Speed after deceleration	550±10rpm	333±10rpm	205±10rpm	300±5%
Rated power	≤4W	≤4W	≤4W	≤6W
Stall current	3A	3A	4A	4A
Rated current	0.3A	0.3A	0.3A	0.5A
Gear reduction ratio	1:19	1:30	1:56	1:40
Encoder type	AB phase incremental Hall encoder	AB phase incremental Hall encoder	AB phase incremental Hall encoder	AB phase incremental Hall encoder
Encoder supply voltage	3.3-5V	3.3-5V	3.3-5V	3.3-5V
Number of magnetic ring lines	11 lines	11 lines	11 lines	11 lines
Interface type	PH2.0 6Pin	PH2.0 6Pin	PH2.0 6Pin	PH2.0 6Pin
Function	With pull-up shaping, the microcontroller can directly read the signal pulse	With pull-up shaping, the microcontroller can directly read the signal pulse	With pull-up shaping, the microcontroller can directly read the signal pulse	With built-in pull-up shaping, the microcontroller can directly read the signal pulse
Weight of single motor	150g±1g	150g±1g	150g±1g	161g±1g

Recommended power supply: 12V. There are four types of 520 motors, and their rated voltage is 12V. When we drive the 520 motor, we can connect a voltage between 11 and 16V, and it is recommended to use a 12V voltage supply.

If you want to distinguish the model of the 520 motor you bought, you can directly look at the label printed on the 520 motor. There is a text of RPM printed on it, and the number in front of RPM corresponds to the number of **speed after deceleration** in the parameter table.

For example, the label of the 520 motor in my hand says 333RPM, so I should pay attention to the parameters in the **MD520Z30_12V** column. Especially the two parameters of **reduction ratio and number of magnetic ring lines**, which may be modified when using the four-way motor driver board.



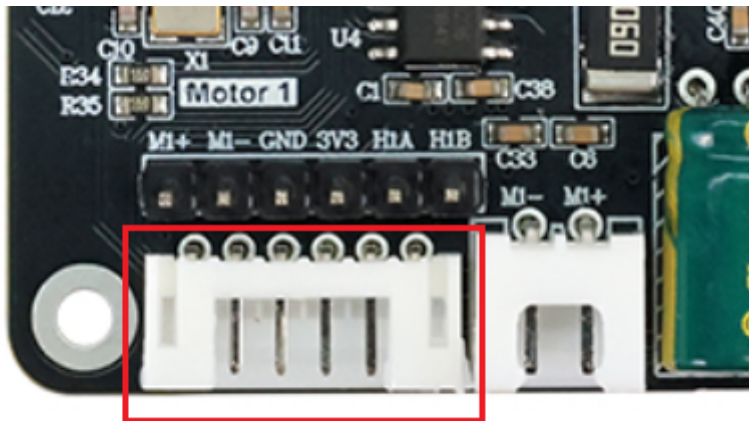
Recommended wiring: The purchased 520 motor will come with two types of wires. Here we recommend that you choose the black PH2.0-6PIN double-head reverse cable, one end connected to the motor, and the other end directly connected to the PH2.0-6PIN encoder motor interface on the four-way motor driver board. This wiring is the most convenient, but it can be found that the A on the motor corresponds to the B phase of the four-way motor driver board. Therefore, when configuring the motor type on the four-way motor driver board, you should choose `$mtype:1#`, the model of the 520 motor.



PH2.0-6pin
double head cable

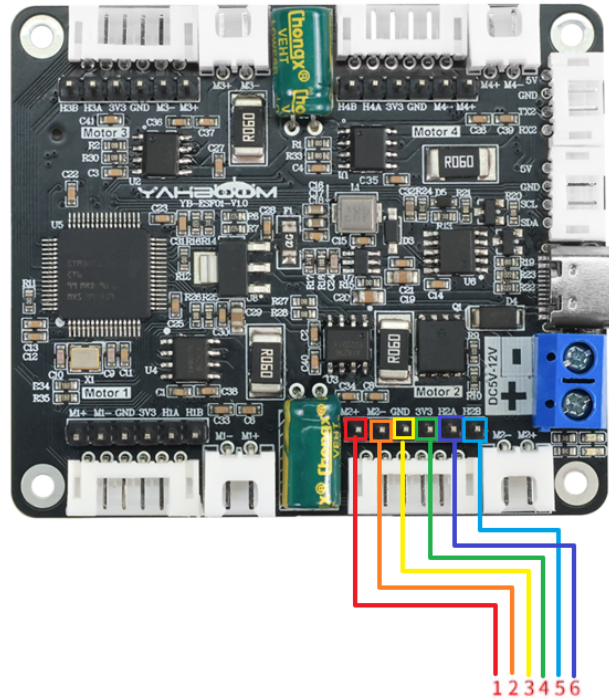


PH2.0-6pin
single head cable



520 motor wiring instructions:

If you use PH2.0-6PIN to Dupont line connection, you can connect according to the picture below. With this connection, the motor phase A will be connected to the four-way motor driver board phase A, and phase B will be connected to phase B. However, when configuring the motor type, you should select `$mtype:2#`, the model of the 310 motor.



- 1: Motor power cable+
- 2: Motor power cable -
- 3: The sensor signal is negative
- 4: The sensor signal is positive 3.3V
- 5: Sensor signal line B phase
- 6: Sensor signal line A phase

